Arm-level models for ORR and median PFS

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Data

Each row is one trial arm (or one non-overlapping stratum of an arm) from the `claude_collected` sheet of `data/raw/step3_stratum_arm_variables.xlsx`: 199 arms from 120 trials. Arms missing any covariate or the outcome are excluded (complete cases), so each model uses fewer arms (see the fit summaries).

Table 1: Covariates $\mathbf{x}_i$ (number of arms missing each)

Symbol	Covariate	Definition	Missing
age_i	Median age (years)	median age	21
F_i	Female proportion	female / n_i	18
T_i	Ever-smoker proportion	ever / (ever + never), known smoking status only	60
E_i	ECOG 0 proportion	ECOG 0 / n_i	43
S_i	Squamous proportion	squamous / n_i	35
G_i	Stage IV proportion	stage IV / stage I–IV, known stage only	55
L_i	First-line (0/1)	1 = first-line incl. maintenance; 0 = pretreated	0
R_i	Maintenance (0/1)	1 = maintenance after induction (subset of first-line)	0
$M_{i,k}$	MOA components (0/1 each)	chemo, ICI, other immune, anti-angio, targeted, other; all 0 = no active drug	0
B_i	Matched biomarker (0/1)	1 = selected on a validated predictive biomarker for a drug in the regimen	0

$\mathbf{x}_i = (\text{age}_i, F_i, T_i, E_i, S_i, G_i, L_i, R_i, M_{i,1}, \ldots, M_{i,6}, B_i)^\top$; the intercept is kept separate. n_i is the arm sample size.

Model 1: logistic models for ORR

Let $Y_i = \text{round}(\text{ORR}_i \cdot n_i)$ be the number of responders in arm i. All three versions share the mean model

$$\text{logit}(p_i) = \alpha_0 + \alpha^\top \mathbf{x}_i,$$

where p_i is the response probability of arm i and $\exp(\alpha_k)$ is the odds ratio of response per unit increase in covariate k. They differ only in the variance of Y_i.

Model 1.1, binomial. $Y_i \sim \text{Binomial}(n_i, p_i)$, with log-likelihood

$$\ell(\alpha_0, \alpha) = \sum_{i=1}^{N} \left[Y_i \lambda_i - n_i \log\left(1 + e^{\lambda_i}\right) \right] + \text{const}, \qquad \lambda_i = \alpha_0 + \alpha^\top \mathbf{x}_i.$$

Model 1.2, quasibinomial. Same mean, $\text{Var}(Y_i) = \phi\, n_i p_i (1 - p_i)$. The point estimates equal Model 1.1; every standard error is multiplied by $\sqrt{\hat{\phi}}$, where $\hat{\phi}$ is the Pearson χ^2 divided by the residual degrees of freedom.

Model 1.3, beta-binomial. $Y_i \mid q_i \sim \text{Binomial}(n_i, q_i)$ with $q_i \sim \text{Beta}(p_i s, (1 - p_i)s)$, so $\text{Var}(Y_i) = n_i p_i (1 - p_i)[1 + (n_i - 1)\rho]$ with $\rho = 1/(1 + s)$. Likelihood

$$L(\alpha_0, \alpha, s) = \prod_{i=1}^{N} \binom{n_i}{Y_i} \frac{B(Y_i + p_i s,\ n_i - Y_i + (1 - p_i)s)}{B(p_i s,\ (1 - p_i)s)},$$

maximized numerically; standard errors from the inverse Hessian.

Fit summary

Table 2: Model 1 fit summary

Model	Arms	Parameters	Dispersion	logLik	AIC
1.1 Binomial	84	16	Pearson $\chi^2/\text{df} = 7.54$	-428.0	887.9
1.2 Quasibinomial	84	17	$\hat{\phi} = 7.54$	–	–
1.3 Beta-binomial	84	17	$\hat{\rho} = 0.040$	-305.7	645.4

The binomial variance clearly fails: the Pearson ratio is far above 1, so Model 1.1 standard errors are too small by about $\sqrt{\hat{\phi}}$. The beta-binomial improves the log-likelihood by 122 units with one extra parameter.

Coefficients

Table 3: Model 1.1, binomial

Term	Estimate	SE	OR (95% CI)	p
Intercept	-7.504	0.677	0.00 (0.00, 0.00)	1.4e-28
Median age (years)	0.050	0.011	1.05 (1.03, 1.07)	5.0e-06
Female proportion	0.925	0.314	2.52 (1.36, 4.67)	0.003
Ever-smoker proportion	-0.981	0.195	0.37 (0.26, 0.55)	4.8e-07
ECOG 0 proportion	-0.449	0.327	0.64 (0.34, 1.21)	0.169
Squamous proportion	0.131	0.123	1.14 (0.90, 1.45)	0.285
Stage IV proportion	2.230	0.444	9.30 (3.90, 22.18)	5.0e-07
First-line	1.178	0.065	3.25 (2.86, 3.69)	2.5e-73
Maintenance	-1.093	0.114	0.34 (0.27, 0.42)	1.2e-21
MOA: chemotherapy	0.863	0.098	2.37 (1.96, 2.87)	1.5e-18
MOA: immune checkpoint inhibitor	0.606	0.068	1.83 (1.60, 2.10)	5.7e-19
MOA: other immune	-0.125	0.111	0.88 (0.71, 1.10)	0.261
MOA: anti-angiogenic	0.678	0.065	1.97 (1.73, 2.24)	2.6e-25
MOA: targeted	0.802	0.094	2.23 (1.86, 2.68)	1.3e-17
MOA: other agent	0.268	0.097	1.31 (1.08, 1.58)	0.006
Matched biomarker	0.400	0.115	1.49 (1.19, 1.87)	4.9e-04

Table 4: Model 1.2, quasibinomial

Term	Estimate	SE	OR (95% CI)	p
Intercept	-7.504	1.857	0.00 (0.00, 0.02)	5.3e-05
Median age (years)	0.050	0.030	1.05 (0.99, 1.11)	0.096
Female proportion	0.925	0.862	2.52 (0.47, 13.66)	0.283
Ever-smoker proportion	-0.981	0.535	0.37 (0.13, 1.07)	0.067
ECOG 0 proportion	-0.449	0.896	0.64 (0.11, 3.70)	0.616
Squamous proportion	0.131	0.336	1.14 (0.59, 2.20)	0.697
Stage IV proportion	2.230	1.218	9.30 (0.85, 101.14)	0.067
First-line	1.178	0.179	3.25 (2.29, 4.61)	4.2e-11
Maintenance	-1.093	0.314	0.34 (0.18, 0.62)	5.0e-04
MOA: chemotherapy	0.863	0.269	2.37 (1.40, 4.02)	0.001
MOA: immune checkpoint inhibitor	0.606	0.187	1.83 (1.27, 2.65)	0.001
MOA: other immune	-0.125	0.306	0.88 (0.48, 1.61)	0.682
MOA: anti-angiogenic	0.678	0.179	1.97 (1.39, 2.80)	1.5e-04
MOA: targeted	0.802	0.258	2.23 (1.35, 3.70)	0.002
MOA: other agent	0.268	0.265	1.31 (0.78, 2.20)	0.313
Matched biomarker	0.400	0.315	1.49 (0.80, 2.76)	0.204

Table 5: Model 1.3, beta-binomial

Term	Estimate	SE	OR (95% CI)	p
Intercept	-7.796	1.616	0.00 (0.00, 0.01)	1.4e-06
Median age (years)	0.045	0.022	1.05 (1.00, 1.09)	0.042
Female proportion	0.179	0.723	1.20 (0.29, 4.93)	0.804
Ever-smoker proportion	-1.001	0.422	0.37 (0.16, 0.84)	0.018
ECOG 0 proportion	0.593	0.624	1.81 (0.53, 6.15)	0.342
Squamous proportion	0.093	0.301	1.10 (0.61, 1.98)	0.756
Stage IV proportion	3.254	1.182	25.88 (2.55, 262.57)	0.006
First-line	1.064	0.160	2.90 (2.12, 3.97)	3.1e-11
Maintenance	-1.221	0.318	0.29 (0.16, 0.55)	1.3e-04
MOA: chemotherapy	0.631	0.223	1.88 (1.21, 2.91)	0.005
MOA: immune checkpoint inhibitor	0.424	0.179	1.53 (1.08, 2.17)	0.018
MOA: other immune	-0.406	0.273	0.67 (0.39, 1.14)	0.137
MOA: anti-angiogenic	0.636	0.154	1.89 (1.40, 2.56)	3.7e-05
MOA: targeted	0.524	0.224	1.69 (1.09, 2.62)	0.019
MOA: other agent	0.104	0.367	1.11 (0.54, 2.28)	0.777
Matched biomarker	0.703	0.266	2.02 (1.20, 3.40)	0.008

Comparison

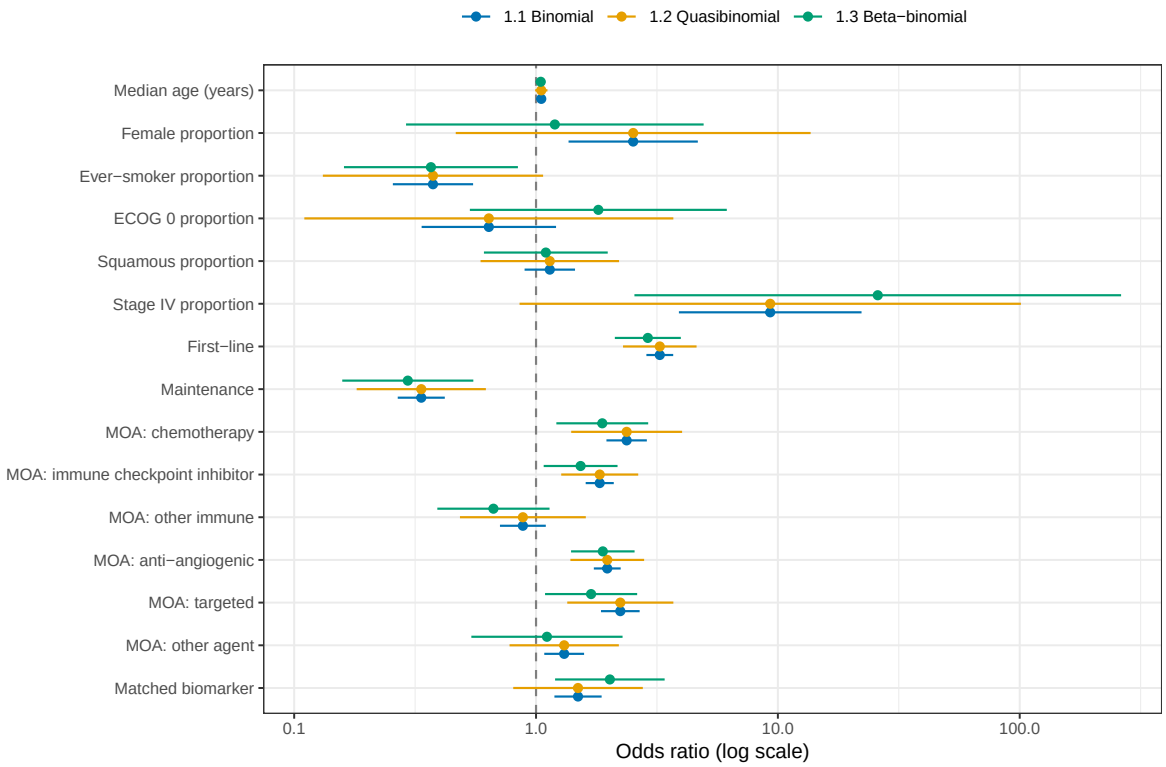


Figure 1: Odds ratios with 95% CIs by model (log scale), intercept omitted.

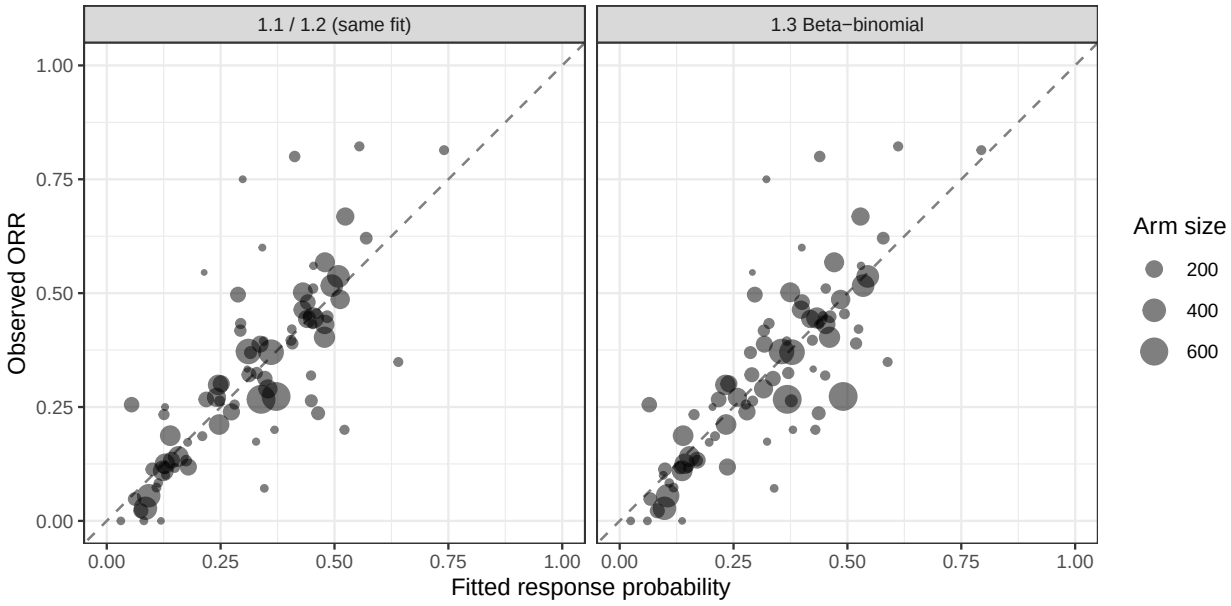


Figure 2: Observed ORR against fitted $\hat{p}_i$; point size is arm size. Points should scatter around the diagonal.

Model 2: log-linear model for median PFS

Let m_i be the reported median PFS (months) of arm i:

$$\log m_i = \theta_0 + \theta_p p_i + \theta^\top \mathbf{x}_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2/n_i),$$

fitted by weighted least squares with weights n_i. $\exp(\theta_k)$ is the ratio of median PFS per unit increase in covariate k. The response probability p_i is not observed and is either

- **observed:** the observed ORR_i (measurement error biases θ_p toward 0), or
- **estimated:** $\hat{p}_i$ from Model 1.1 (two-stage); standard errors and CIs from 1000 bootstrap resamples of arms.

Observed ORR

Table 6: Model 2 with observed ORR (80 arms, residual SE 2.07)

Term	Estimate	SE	Median ratio (95% CI)	p
Intercept	-0.658	0.614	0.52 (0.15, 1.76)	0.288
Response probability p_i	1.343	0.218	3.83 (2.48, 5.93)	5.9e-08
Median age (years)	0.030	0.010	1.03 (1.01, 1.05)	0.005
Female proportion	-0.050	0.291	0.95 (0.53, 1.70)	0.864
Ever-smoker proportion	-0.199	0.194	0.82 (0.56, 1.21)	0.309
ECOG 0 proportion	0.099	0.302	1.10 (0.60, 2.02)	0.743
Squamous proportion	-0.149	0.122	0.86 (0.67, 1.10)	0.228
Stage IV proportion	-0.294	0.397	0.75 (0.34, 1.65)	0.462
First-line	0.156	0.078	1.17 (1.00, 1.37)	0.049
Maintenance	0.094	0.101	1.10 (0.90, 1.34)	0.354
MOA: chemotherapy	0.420	0.091	1.52 (1.27, 1.82)	1.9e-05
MOA: immune checkpoint inhibitor	0.141	0.070	1.15 (1.00, 1.33)	0.050
MOA: other immune	-0.029	0.107	0.97 (0.78, 1.20)	0.786
MOA: anti-angiogenic	0.160	0.066	1.17 (1.03, 1.34)	0.018
MOA: targeted	0.090	0.092	1.09 (0.91, 1.31)	0.334
MOA: other agent	0.008	0.115	1.01 (0.80, 1.27)	0.947
Matched biomarker	0.419	0.105	1.52 (1.23, 1.88)	1.8e-04

Estimated $\hat{p}_i$ (two-stage)

Table 7: Model 2 with $\hat{p}_i$ from Model 1.1 (80 arms)

Term	Estimate	SE	Median ratio (bootstrap 95% CI)	p
Intercept	-1.413	1.936	0.24 (0.00, 5.37)	0.466
Response probability p_i	0.308	1.783	1.36 (0.03, 32.46)	0.863
Median age (years)	0.040	0.027	1.04 (1.00, 1.11)	0.138
Female proportion	0.030	0.521	1.03 (0.29, 2.46)	0.955
Ever-smoker proportion	-0.402	0.492	0.67 (0.19, 1.50)	0.414
ECOG 0 proportion	0.067	0.511	1.07 (0.50, 3.51)	0.896
Squamous proportion	-0.151	0.212	0.86 (0.55, 1.30)	0.478
Stage IV proportion	-0.084	1.023	0.92 (0.14, 7.47)	0.935
First-line	0.375	0.382	1.45 (0.68, 3.18)	0.327
Maintenance	-0.101	0.349	0.90 (0.40, 1.59)	0.772
MOA: chemotherapy	0.583	0.299	1.79 (1.06, 3.65)	0.051
MOA: immune checkpoint inhibitor	0.262	0.244	1.30 (0.90, 2.38)	0.284
MOA: other immune	-0.049	0.224	0.95 (0.50, 1.24)	0.826
MOA: anti-angiogenic	0.290	0.250	1.34 (0.88, 2.44)	0.246
MOA: targeted	0.264	0.351	1.30 (0.71, 2.99)	0.453
MOA: other agent	0.074	0.152	1.08 (0.84, 1.58)	0.626
Matched biomarker	0.488	0.415	1.63 (1.13, 5.87)	0.240

θ_p is weakly identified here: $\hat{p}_i = \text{logit}^{-1}(\hat{\alpha}_0 + \hat{\alpha}^\top \mathbf{x}_i)$ is a function of the same $\mathbf{x}_i$ that enters Model 2, so θ_p is separated from θ only by the curvature of the logit. The wide interval for θ_p reflects this, not a weak ORR–PFS association. A joint model with latent p_i informed by the observed responders is the principled alternative.

Notes

- ORR is reported as a proportion; Y_i is reconstructed by rounding $\text{ORR}_i \cdot n_i$.
- Arms from the same trial are treated as independent. A trial-level random effect would account for their correlation.
- Overlapping or post-treatment subgroups (NCT01523587 mainland China subgroup, NCT03850444 KEYNOTE-042 China study, NCT01774721 skin-AE subgroups, NCT01038661 maintenance-phase arms) were moved to the `irrelevant_arms` sheet before fitting.